Harik Sodhi

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AWARDS & HONOURS

- **Silver Medal, International Physics Olympiad** (2025) – 49th/420 competitors from 89 countries; highest-ranked competitor from Western Europe.
- **United Kingdom International Physics Olympiad Team Member** (2025) – Selected after Gold awards in BPhO Round 2 and BAAO Round 2.
- **Winner, Berkeley Math Tournament Online Power Round** (2024) – First place in a proof-based competition on graph theory, cryptography, and computational complexity.
- **Ritangle National Champion** (2023) – Team Captain; 1st/1,200 teams in a national combinatorial optimisation challenge, developing and proving an optimal solution to a highly constrained scheduling problem with hundreds of interacting variables.

EDUCATION

University of Oxford (Christ Church) | *MEng Engineering Science* — **Oct 2025 – Jun 2029**

- First-year: 93% labs, 92% internal exams

EXPERIENCE

Oxford Alpha Fund — Oxford

Quant Analyst / Director — *Jan 2026 – Present*

- Selected through a competitive quantitative research and presentation process as a first-year undergraduate.
- Researched volatility risk premia in short-dated Bitcoin options using historical Deribit/Coinbase data; developed and backtested systematic trading strategies achieving a Sharpe ratio of 1.18 and a 70% win rate. [**Code**]
- Serve on the general committee, contributing to the Walton Edge newsletter, sponsor liaison, and candidate evaluation.

Optiver Trading Academy — Oxford

Participant — *Oct 2025 – Nov 2025*

- Selected for Optiver's competitive Trading Academy programme as a first-year undergraduate.
- Traded ETFs and futures on Optiver's proprietary simulation platform, applying market-making, delta-hedging, and risk-management principles under realistic market conditions.

LEADERSHIP & ACTIVITIES

VEX Robotics & REC Foundation — London / Remote

Global Student Advisory Board Member & Team Captain — *May 2021 – Mar 2024*

- Selected as one of 12 members globally to directly advise the CEO and executive leadership team on the strategic direction, international expansion, and accessibility of the REC Foundation—the governing body of the world's largest robotics competition spanning 2,400+ teams from 60+ countries and major corporate partnerships (e.g., NASA).
- Served as Team Captain, Lead Programmer, and Lead Designer; managed team operations and spearheaded end-to-end development of competitive robots.
- Developed autonomous routines and driver-assist systems in C++ using PID control and designed competition mechanisms in CAD.

PROJECTS

Market Making Simulator — Nov 2025

- Developed a multi-agent market simulation modelling market makers, noise traders, and informed traders.
- Analysed the effects of participant composition on spread formation, adverse selection, and trading profitability.
- Implemented strategy evaluation and performance analysis tools for comparing trading behaviour. [**GitHub**]

Rocket Lander — Apr 2026

- Designed and developed a MATLAB simulation of autonomous rocket landing dynamics from first principles, incorporating variable-mass propulsion, atmospheric drag, and gravity.
- Implemented Runge-Kutta numerical integration, a scratch-built PID control system, and constrained optimisation techniques to achieve stable autonomous touchdowns. [**GitHub**]

SKILLS

- **Programming:** Python, C++, MATLAB, LaTeX
- **Quantitative:** Statistical Modelling, Backtesting, Optimisation, Stochastic Simulation, Volatility Analysis, Numerical Methods
- **Libraries:** NumPy, Pandas, SciPy, PyTorch, Scikit-learn, Matplotlib
- **Tools:** Linux, Git, GitHub, SolidWorks